Verification and Progress on Purity of Pure

A machine-checked report on Goodman's project notes

27 July 2026; updated 31 July 2026

Purpose. This report records what we have verified in Jeremy Goodman's July 2026 project notes on the consistency of Purity of Pure with Bacon's logical combinatorialism. It has two aims. First, it separates the claims that have been verified exactly from claims that require qualifications, repairs, or further formulation. Second, it states the new theorems proved during the formalization that genuinely narrow the open consistency problem.

Bottom line. Goodman's central consistency question remains open. Isabelle nevertheless settles the determinate subsidiary claims in the notes: each is proved, refuted and repaired, or assigned the explicit qualification under which it is true. In particular, the formalization identifies and repairs a modal gap in T3 and a false countability argument in M5; verifies the four T6 contradiction routes and the T9 cardinal dichotomy from their explicit premises; completes the rebuilt M5 model containing the displayed exotic operator; and completes the denotational step in Bacon's footnote-59 argument. Over Bacon's exact recursively restricted carriers, the displayed footnote-59 term denotes the required diagonal, with identity interpreted at the appropriate higher types. Consequently no interpretation in that semantic framework can validate PP together with the PP-free M1 background and QSS. This rules out completing that exact-carrier interpretation while retaining the PP-free M1 background (including Purity of Fun), QSS, and unique fundamentality. The same argument has now been proved for arbitrary compositional CEV+ Henkin models: no such model validates full QLN, PP, and Purity of Fun. This still does not settle Goodman's question, because Purity of Fun is deliberately absent from the central axiom sets.

A separate source-fidelity audit confirms the object-language encodings, including every orientation-sensitive clause in T6. It also identifies three qualifications that should be kept explicit. Goodman's PC is sufficient for his T9 counting argument, but it is weaker than Bacon's Melianism premise for the argument concerning PP. The infinite-disjunction argument appears in Bacon's main text on p. 27 with notes 57–58, not in note 60. Finally, Bacon leaves two claims required by later arguments in the appendix unproved: Theorem 10.1 and the assertion that the resulting interpretation validates QLN. Isabelle now supplies both missing verifications in the unique-fundamental-proposition specialization relevant to Goodman's question. In that setting the only nonvacuous QLN instances are zeroary and unary. The zeroary Exhaustion direction follows from the root classification of closed logical propositions; the unary Exhaustion direction follows from invariance under cone restriction; and

Recombination follows from the generic fundamental proposition. Bacon's more general suggestion concerning arbitrary individual types or several pairwise distinct fundamental entities remains outside this verification.

The formalization also proves that Recombination together with zeroary Exhaustion and unique proposition-level fundamentality supplies the fun' witness used in T6. Hence T6 has the following exact semantic consequence: any model of the repaired background theory plus PP must falsify global $\mathsf{L2}$ or global TU. That consequence sharply constrains a positive answer, but it does not decide whether such a model exists. We have also settled Goodman's proposed calibration of $\mathsf{L2}$ in Bacon's appendix model: a closed logical operator Z refutes $\mathsf{L2}$ when the pure unary operators are exactly the denotations of closed terms containing only logical vocabulary. Isabelle proves the required composition closure and constructs the fun' proposition relative to this stock. Finally, we have now carried Goodman's second and third suggested attacks farther. A type-correct Pure Completeness construction, together with $\mathsf{L2}$ and the standard action of pure reversible operators on kinds, implies that there are infinitely many kinds. T9 then forces $2^{|K|} \leq |G|$. On the granularity side, a reversible operator is uniformly truth-preserving or uniformly truth-flipping whenever it sends propositions with the same truth value to propositions with the same truth value. Under QLN this conclusion is equivalent to a single noncontingency condition at a fundamental proposition. These are conditional theorems: the needed $\mathsf{L2}$, classifier realization, and noncontingency premises have not been derived from the central CEV+ stock with PP. This is a substantive result about the notes, although it does not construct the missing interpretation of Pure and therefore does not answer the consistency question. Enlarging the pure stock changes the meaning of fun', so no unrestricted transfer to enlarged stocks is claimed.

1 Formal setting and verification standard

The formal development represents Bacon's typed language and proof system directly in Isabelle/HOL. It includes Bacon's system H, classical and Boolean identity principles, the Rule of Equivalence, vector Equivalence, the definition of necessity by identity with truth, and the additional purity/fundamentality vocabulary used in Goodman's notes. The development distinguishes object-language derivability from validity in a particular model. This matters throughout: a semantic argument valid in Bacon's appendix model is not thereby a derivation in the theory, and an Isabelle theorem with explicit premises does not show that those premises hold in the intended model. More exactly, the base rules reproduce Bacon–Dorr's displayed system H^- and add the individual Existence instance

$$\exists x_{\mathsf{Ind}}\,(x = x),$$

which Bacon and Dorr use to recover H in the simple type system at issue. The exact HOL–ZF carrier at type Ind is a singleton, so Isabelle also verifies this instance semantically and propagates it through the soundness proofs for H, Classicism, CE, and CEV. Bacon–Dorr prove that Equivalence is admissible over Classicism. The Isabelle organization takes the corresponding rules as primitive in its CE and CEV presentations; this is a presentational convenience, not an attributed hierarchy of proof-theoretic strength. The appendix-model, rebuilt-M5, exact-M1, and exact-L2 results use a HOL-ZF interpretation whose carrier satisfies axiomatized ZFC; they are therefore relative semantic theorems in that setting, not pure-HOL consistency certificates for ZFC. The development contains an explicit soundness theorem for the CEV+ axiom extension. Its assumptions require soundness of the underlying CEV calculus, soundness of vector Equivalence, and global validity of every added axiom. The exact appendix interpretation discharges the first two obligations. Consequently, showing that an interpretation validates the complete added stock would genuinely establish CEV+ consistency; that validity obligation is the open problem, not an assumption hidden in the soundness theorem.

1.1 Exact appendix-model metatheory

The source-faithful HOL–ZF development now completes the following four tasks over Bacon's recursively defined carriers.

Task	Verified result
1	Isabelle proves Bacon's arbitrary-signature Theorem 10.1 throughout the t-fragment, comprising all types built from t alone, and supplies the branch-gluing proof Bacon omits. Signatures involving e-containing types remain outside scope, as Bacon explicitly defers that extension.
2	Soundness is proved for H, Classicism, CE, and CEV, including the individual Existence instance and vector Equivalence for an arbitrary finite vector of argument types.
3	The countable enumeration of consistent sentences and Bacon's gluing construction are proved over the exact carriers.

Task	Verified result
4	One glued model represents consistency and the complete frame theory for the fixed proposition-generated signature. This is Bacon's semantic frame-theory representation theorem, not proof-theoretic completeness of H.

The dedicated Goodman-notes audit checks 167 theorem objects supporting the central object-language and model-theoretic claims below, and fresh builds pass. Separate maintained sessions check the modal-and-quantified fragment models and the later finite-fragment constructions. None of these proofs contains an admitted lemma, an untrusted proof step, or an undischarged assumption beyond the assumptions stated in the theorem itself. The type restriction carried by the general cardinal theorems in T9 is part of their formally checked statement, not a further mathematical premise. This kernel check certifies proof-object hygiene: no oracles, no dangling hypotheses, and no unsolved unification pairs. It does not by itself show that a theorem's stated assumptions are weak, source-faithful, or nonvacuous. Those questions were checked separately by comparing the statements and their uses with Bacon's, Dorr's, and Goodman's texts.

The semantic evaluation used below interprets identity at proposition type, the type of unary operators, and the next classifier type in their respective domains. In particular, identity between unary operators is pointwise identity over the proposition domain; it is not obtained by first treating higher-type values as propositions. This correction to an auxiliary evaluator preserves all earlier verified results and is essential to the exact M1 calculation.

2 Background definitions and two important distinctions

At proposition type, Goodman's definable surrogate for fundamentality is

$$\mathsf{fun}'(p) \quad \stackrel{\text{def}}{=} \quad \forall X\, Y\big((\mathsf{Pure}(X)\ \&\ \mathsf{Pure}(Y)\ \&\ Xp = Yp) \to X = Y\big).$$

Let G be the pure reversible unary operators, and let

$$X \approx Y \quad \stackrel{\text{def}}{=} \quad \exists Z \in G\ (X = Y \circ Z).$$

Weak $\mathsf{L2}$ says that if X and Y are pure, p and q are fun', and $Xp = Yq$, then $X \approx Y$.

Two distinctions uncovered by the formalization are central.

The scope of QLN. Goodman's general formulation requires the fundamental arguments of a QLN instance to be pairwise distinct. The present problem stipulates exactly one fundamental entity, which is a proposition. Hence every instance of arity at least two is vacuous. The zeroary and unary instances are therefore the complete substantive QLN package for this problem; no higher-arity instance has been silently omitted. This last reduction is a metatheoretic observation recorded in the documentation; it is not a separate object-language Isabelle theorem.

Recombination versus Exhaustion. Unary Recombination proves a pointwise modal identity principle. It does not, by itself, permit necessitating the universal closure required by the usual QSS derivation. Zeroary Exhaustion supplies exactly that missing step. With unique proposition-level fundamentality, the repaired package derives the existence of a fun' proposition.

The scope of Modalized Functionality. Bacon–Dorr's Intensionality argument, as formalized in CEV, proves Modalized Functionality for operators whose result type is proposition:

$$X, Y : \sigma \to t.$$

The argument type σ is arbitrary. The broader two-type formula for $X, Y : \sigma \to \tau$, with arbitrary result type τ, remains a separate schema and a direct model obligation. The T2–T8 arguments formalized here use the proposition-valued instances and are unaffected by this distinction. What is not justified is the earlier suggestion that the entire arbitrary-result schema is already redundant over bare CEV.

Conditional model-theoretic results. A formally verified theorem may still have explicit premises concerning Bacon's function domains, invariance, or QSS. We therefore distinguish a verified conditional theorem from a proof that all of its premises hold in Bacon's particular appendix model.

3 Goodman's L2 principles

Goodman's July notes state one principle bearing the label L, in two forms. Weak $\mathsf{L2}$ is

(L2) $$\mathsf{Pure}(X) \,\&\, \mathsf{Pure}(Y) \,\&\, \mathsf{fun}'(p) \,\&\, \mathsf{fun}'(q) \,\&\, Xp = Yq \quad\Longrightarrow\quad X \approx Y.$$

The strong form requires the same reversible operator to witness both the relation between the operators and the relation between their inputs:
(strong L2)

$$\mathsf{Pure}(X) \,\&\, \mathsf{Pure}(Y) \,\&\, \mathsf{fun}'(p) \,\&\, \mathsf{fun}'(q) \,\&\, Xp = Yq \quad \Longrightarrow \quad \exists Z \in G\,(X = Y \circ Z \,\&\, q = Zp).$$

The notes contain no separate principle labeled $L1$. Accordingly, the plural "L principles" below refers to weak and strong L2.

3.1 Object-language verification

Both forms have been encoded as closed, well-typed object-language formulas. Isabelle verifies all four uses made in T6:

$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{Inv} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{WI} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \exists\mathsf{fun}' + \mathsf{L2} + \mathsf{TU} \vdash \bot,$$
$$T_0 + \mathsf{PP} + \text{strong } \mathsf{L2} + \mathsf{RS} \vdash \bot.$$

The repaired Recombination–Exhaustion bridge derives the required $\exists\mathsf{fun}'$ premise from Recombination, zeroary Exhaustion, and unique proposition-level fundamentality. This is a repaired route from Goodman's official fundamental-proposition formulation. The four original T6 stocks instead assume $\exists\mathsf{fun}'$ directly and contain no QLN principle at all; their advertised compatibility with a Recombination-only setting is therefore correct. Isabelle also verifies the uses of L2 in T7a and T8, including uniqueness of kind in T8. T9 is a separate conditional cardinal theorem derived from its explicit PC-injectivity, representation, and orbit-fibre coding assumptions; it does not use L2.

3.2 Refutation of L2 in Bacon's exact appendix model

Goodman's first suggested attack asks whether L2 holds in Bacon's appendix model when the pure unary operators are exactly the denotations of closed terms containing only logical vocabulary. Isabelle answers this question negatively on Bacon's actual finite-natural-word action.

In the prefix presentation, define

$$\hat{Z}(P) = \{\, w : (\exists n\; P(w^\frown\langle n\rangle))\,\&\,(\exists n\; \neg P(w^\frown\langle n\rangle))\,\}.$$

Thus $\hat{Z}(P)$ holds at w iff P is true at some immediate successor of w and false at some immediate successor of w. Reversal intertwines this prefix presentation with Bacon's right-division action. In Bacon's raw-word presentation the same operator is

$$Z(P) = \{\, i : (\exists n\ \langle n\rangle^\frown i \in P)\ \&\ (\exists n\ \langle n\rangle^\frown i \notin P)\,\}.$$

Isabelle proves that a closed term containing only logical vocabulary has exactly this denotation. Hence Z belongs to Bacon's stock of unary operators denoted by closed logical terms. It further proves:

(i) Z is surjective;
(ii) $Z(P) = Z(\neg P)$, so Z is not injective;
(iii) Z is right-cancellative on that stock;
(iv) $Z \notin G$.

Surjectivity is witnessed explicitly by

$$P_S = \{\, \langle 0\rangle^\frown i : i \in S\,\}, \qquad Z(P_S) = S.$$

This stock is countable and composition-closed. A generic-separation argument supplies a proposition p satisfying fun' relative to that stock. Clause (iii) is equivalent to saying that Zp also satisfies fun', and

$$\mathsf{id}(Zp) = Zp = Z(p).$$

If L2 held, identity and Z would therefore be of the same kind. A reversible witness for that claim would supply a left inverse for Z, making Z injective. Clause (ii) rules this out. Consequently both weak and strong L2 fail when the pure unary operators are precisely the denotations of closed logical terms in Bacon's model.

3.3 What this advances

This settles the first open problem in Goodman's list for Bacon's exact interpretation: L2 is false when purity is interpreted by the complete stock of closed-logical denotations. The operator Z is a verified nonreversible closed logical operator that nevertheless preserves the distinctions relevant to fun'. The formula above also corrects the earlier binary-tree calculation: Bacon's frame has one immediate successor for each natural number, so its exact operator tests variation among all immediate successors rather than comparing two distinguished children.

The result does not prove the consistency of PP. In the exact-carrier closed-logical interpretation, PP at the unary-operator type is equivalent to membership of the unary-stock classifier in the next closed-logical stock. That membership has not been proved or refuted unconditionally. Nor does the theorem show that every larger interpretation of the pure unary operators still falsifies L2: right-cancellativity and fun' must be recalculated relative to the enlarged stock. What the result establishes is the exact calibration Goodman requested: closed logical definability by itself does not make L2 valid in Bacon's appendix model.

3.4 Automated proof search for L2 from PP

We also used Vampire to test whether a finite representation of $T_0 + \mathsf{PP} + \exists\mathsf{fun}'$ yields weak L2. The first experiment presented the entire represented stock together with the denial of L2 as one higher-order refutation problem. It returned no proof after one hour.

We then divided the proposed derivation into separate intermediate claims. Suppose that X and Y are pure, p and q are fun', and $Xp = Yq$. Vampire proves that X and Y then agree on all equations obtained by composition on the left by pure operators:

$$(*)\qquad \forall A\,B\,(\mathsf{Pure}(A)\,\&\,\mathsf{Pure}(B) \implies (A \circ X = B \circ X \;\leftrightarrow\; A \circ Y = B \circ Y)).$$

It also proves each of the following conditional routes to L2:

1. if pure X and Y satisfying $(*)$ must satisfy $X \approx Y$, then L2;
2. if every two fun' propositions p, q satisfy $q = Zp$ for some $Z \in G$, then L2;
3. even the weaker condition that, for every pure Y and every two fun' propositions p, q, there is some $Z \in G$ such that $Yq = Y(Zp)$, implies L2.

The remaining question is whether $T_0 + \mathsf{PP} + \exists\mathsf{fun}'$ proves any one of these three sufficient conditions. We tested each condition separately, first with genuine higher-order function types and then in a multisorted first-order representation in which unary operators are reified as objects. With a 60-second, six-core search for each conjecture, all six runs timed out. None returned either a proof or a countermodel.

These experiments establish no nonderivability result. Their positive value is diagnostic: Vampire rapidly proves the algebraic steps from each proposed intermediate condition to L2, but it does not presently obtain any of those conditions from PP. Thus the automated search isolates the same mathematical gap as the formal analysis: a negative argument along this route requires a substantive new consequence of PP, rather than merely a longer search through the already identified composition and cancellation steps.

4 Audit of the object-language theorems

Item	Status	Result
T1	Verified	Under zeroary Exhaustion, every pure proposition is truth or falsity. Consequently the biconditional operators are identity and negation, and WI collapses to Inv.
T2a	Verified	fun' is closed under every pure reversible operator, hence under negation.
T2b	Verified	Truth and falsity are not fun'. A fun' witness is distinct from truth, falsity, and its negation.
T2c	Verified, stronger	Every pure proposition is possibly identical to a fun' witness. The proof does not use PP.
T2d	Verified, stronger	Possible purity of the witness follows without Persistence, although Goodman's displayed route uses it.
T2e	Verified	The proposition that the witness is noncontingent is false but possible.
T2f	Verified conditionally	The six displayed propositions are pairwise distinct under the explicit $\mathsf{fun}'(r)$ antecedent. The antecedent is essential.
T3	Advertised form unproved; repaired forms verified	The advertised proof has a modal gap: it reaches possible identity where identity is required. The theorem follows after adding zeroary Exhaustion, or from a weaker pure-identity rigidity principle. A two-world modal abstraction verifies that the advertised intermediate principles do not by themselves close the gap. This is a countermodel to the advertised modal skeleton, not a full CEV+ countermodel, so it does not establish nonderivability from the complete theory.
T4	Verified, stronger	For each closed pure $C : t \to (t \to t)$, $\sim\mathsf{fun}'_{t\to t}(C(r))$. The proof uses only the purity schema and application closure; neither PP nor $\mathsf{fun}'(r)$ is needed.
T5	Verified conditionally	The assumptions displayed in the notes, together with $\mathsf{fun}'(r)$, yield a fun' proposition distinct from r and $\sim r$. This does not establish consistency of the antecedent.

Item	Status	Result
T6	Verified	All four contradiction routes are formalized: L2+Inv, L2+WI, L2+TU, and strong-L2+RS. Recombination, the proposition-type instance of Exhaustion, and unique fundamentality derive, rather than assume, the required fun' witness in the repaired stocks. The original stocks assume $\exists\mathsf{fun}'$ and are QLN-free. The advertised T6-WI master claim is derived directly from its stated stock, independently of explosion.
T7a	Verified	For the liar family, the identity-indexed member is false, while some member indexed by a pure reversible operator is true.
T7b	Underspecified	The notes do not give the kind-level diagonal term, its binders, or precise definitions of "fixed" and "shifted" point. There is no unique theorem to encode.
T8	Verified	The five base kinds are pairwise distinct; L2 gives kind uniqueness; and the assumptions displayed in the notes prove 31 pairwise-distinct pure unary operators and 31 pairwise-distinct propositions.

Item	Status	Result
T9	Counting verified; conditional infinitude verified	The PC specification makes the map from sets of kinds to pure operators injective. Representation of the members of each kind by members of G gives an injective code of that kind into G. These results imply $$\lvert \mathcal{P}(K) \rvert \leq \lvert K \times G \rvert.$$ Classical cardinal arithmetic then gives $$K \text{ finite} \quad \vee \quad \lvert \mathcal{P}(K) \rvert \leq \lvert G \rvert.$$ This is the correct use of PC. It does not identify PC with Bacon's stronger Melianism thesis about purity. A type-correct semantic realization of the Pure Completeness selector, together with the required instance of L2 and the standard action of pure reversible operators on kinds, excludes the finite disjunct and therefore yields infinitely many kinds. Discharging those semantic premises from Goodman's full object-language theory remains separate.

5 Audit of the model-theoretic claims

Item	Status	Result
M1	Exact denotation and general Henkin exclusion, with qualification	At proposition type, the pure propositions in Bacon's model are exactly the two invariant propositions, and the operator testing for membership in that class is the noncontingency operator. The exact footnote-59 liar is typed, beta-reduced, and proved pure from PP, Purity of Fun, and application closure. Isabelle now also computes its denotation in Bacon's exact tree semantics. It is precisely Goodman's displayed diagonal, with proposition and operator identity interpreted by the appropriate local equivalence relations. The object-language formula QSS is evaluated in the same semantics, and the resulting diagonal argument is proved without an assumed semantic diagonal premise. The calculation has also been abstracted to arbitrary compositional CEV+ Henkin models. No such model validates full QLN, PP, and Purity of Fun. At the next type, the infinitary join still exists in Bacon's full function domain and has exactly the pure unary operators as its extension; PP holds there exactly when that classifier is itself in the next pure stock. The full function domain supplies the classifier, but the footnote-59 theorem shows that it cannot be pure while the other M1 assumptions and QSS are retained. Thus Isabelle proves the exact PP/classifier equivalence and a conditional nonpurity result; it does not establish Goodman's unconditional assertion that the infinitary join is impure. The qualification is essential: Goodman's central axiom sets omit Purity of Fun, which the proof that the footnote-59 operator is pure uses.
M2	Verified	Goodman's construction $T \mapsto G_T$ is a bijection between arbitrary sets of propositions and invariant unary operators. Cantor's theorem yields more invariant operators than propositions, so evaluation at one fundamental proposition cannot be injective on all invariants. The invariance reading of purity therefore violates QSS.

Item	Status	Result
M3	Verified relative to the pure operators	Relative to the chosen stock of pure operators, fun' is equivalent to freeness against nonzero pure laws. Such propositions have both extreme views. A countable Boolean stock has a free generator, while the resulting fun' class is topologically meager in the product topology.
M4	Conditional model-theoretic theorem	Preimage heredity is proved. The lifted-branch witness is fun' and outside the reversible orbit under explicit assumptions: necessitated QSS, function-space membership and invariance of every operator in the pure stock, and presence of identity, zero, and one. The theorem has not yet been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. The multiple-fundamental "wide Fun" argument remains prose-level.
M5, construction	Verified and repaired	The operator $f = G_T$ obtained by exchanging s and s' is an invariant involution violating TU, WI, and Inv. The notes' subsequent countability argument is false: both the orbit and the displayed world-indexed family are countable. We replace it with a uniform diagonal construction. For every candidate R, it produces a two-element pair outside every view of R, whose proper views are all empty or universal. The induced invariant nonidentity transposition fixes R. If it and identity are both included among the pure operators, R ceases to be fun', proving the corrected pre-rebuild QSS obstruction.

Item	Status	Result
M5, rebuild	Verified in a secondary comparison model	Isabelle constructs the least application-closed pure stock containing every closed logical denotation and the displayed f, proves that f is a pure self-inverse operator in the modalized function domain, and rebuilds a fundamental proposition that supplies Recombination and fun'-separation at every world. This Boolean-tree comparison interpretation validates Bacon's Recombination background. It does not establish PP; that would require the classifier of the enlarged pure stock itself to belong to the next pure stock.
M5, collision	Verified; unrestricted generalization refuted	The displayed collision calculation is verified: the closed operator $$\lambda p\,(p \leftrightarrow \mathsf{NC}(p))$$ takes the same value at $\mathsf{NC}(r)$ and truth, although $\mathsf{NC}(r) \neq \mathsf{truth}$, whenever $\mathsf{fun}'(r)$. Extending this collision argument uniformly to merely existentially given invertibles is impossible. Isabelle proves in $\mathsf{CEV}+$ that every operator with an existentially supplied two-sided inverse is injective; in particular, the displayed collision operator is not reversible. Conversely, the rebuilt model's exotic operator is a reversible, non-truth-uniform bijection and has no collision. A PP-specific collision argument would therefore require a further condition constraining the action of the inverse witness.
M6	Conditional model-theoretic theorem	Under the same explicit assumptions—necessitated QSS, membership of every pure operator in Bacon's function domain, and invariance—distinct substitutions are separated by a fun' proposition. With identity, zero, and one, a strict-inclusion fun' pair blocks the indicated joint assignment. Single-coordinate arbitrary realization fails independently by countability. These results have not all been instantiated for Bacon's stock of operators denoted by closed terms containing only logical vocabulary.

Item	Status	Result
M7	Abstract theorem verified; exact instance open	In the secondary Boolean-tree comparison model, the Tarski diagonal lies outside the range of the enumerated closed terms. For arbitrary invariant operators of Goodman's form G_T, every proposition is reachable from r if and only if the orbit map $$i \longmapsto i \cdot r$$ is injective. Whether Bacon's glued r has this property is construction-sensitive and open.
10.1	Verified for arbitrary signatures in the t-fragment	Isabelle proves Bacon's arbitrary-signature theorem throughout the fragment of types built from t alone and supplies the branch-gluing proof Bacon omits. Signatures involving e-containing types remain outside scope, as Bacon explicitly defers that extension. The verified scope includes Goodman's language with its single propositional constant. The verified M5 rebuild no longer depends on Theorem 10.1.

6 Corrections and clarifications to the notes

6.1 Recombination does not by itself yield QSS

The notes say that QSS follows from Recombination alone. The formal proof shows that this is false as stated. Recombination applied to the pure operator

$$\lambda p\,(Xp = Yp)$$

yields the unboxed universal identity

$$\forall p\,(Xp = Yp).$$

Modalized Functionality requires its necessitation. Since the derivation occurs under assumptions, admissible Necessitation cannot supply that step. Zeroary Exhaustion does: it makes the pure universal proposition necessary, after which Modalized Functionality yields

$X = Y$.

This correction affects every route from a fundamental proposition to fun', including the corresponding reading of T2 and the repaired T6 stocks. It does not affect the original T6 contradictions, which assume $\exists\mathsf{fun}'$ directly and are QLN-free.

6.2 The modal gap in T3

The advertised argument uses necessitated QSS and Persistence to move from a possible fundamental role to an injectivity claim. The formal derivation reaches a statement of the form

$$\Diamond(Y = Z),$$

where the desired conclusion requires $Y = Z$. No rule of H licenses that transition. The gap can be repaired in either of two principled ways.

1. Zeroary Exhaustion collapses pure propositions to truth and falsity, which supplies the required rigidity.
2. One may assume only the weaker principle that possible identity between pure operators implies identity.

The unrestricted version of the second principle is false. Thus the repair must remain restricted to the pure case.

6.3 PC and Bacon's Melianism premise

Goodman's PC says that there is a pure property coextensive with the pure entities. Bacon's argument for PP on p. 27 assumes the stronger Melianism identity

$$\mathsf{Pure}_\sigma = \lambda x\,(x = a_1 \vee \cdots \vee x = a_n),$$

with an infinitary analogue when the pure entities are infinite. Bacon's note 58 explicitly observes that the argument without Melianism establishes only coextensiveness with a pure property, not identity with Pure_σ. Accordingly, PC is sufficient for Goodman's T9 counting argument but should not be presented as a reconstruction of Bacon's argument for PP.

6.4 The infinite disjunction and the footnote citation

Goodman's discussion attributes the infinitary-disjunction argument to Bacon's note 60. The argument is instead in the main text on p. 27, with notes 57–58; note 60 merely directs the reader to the appendix and further work. The machine-checked result is also weaker than Goodman's unconditional nonpurity claim. The infinitary join exists in Bacon's full unary function domain and has exactly the pure unary operators as its extension. It is pure exactly when the relevant instance of PP holds. Its nonpurity follows only with the additional diagonal and QSS assumptions described above.

6.5 The two omitted verifications in Bacon's appendix

Bacon does not prove Theorem 10.1 in *Logical Combinatorialism*; Isabelle supplies the missing proof for the fragment to which $A(M)$ applies. Bacon also says, without proof, that the resulting interpretation validates QLN. Isabelle now verifies that assertion for the specialization used in Goodman's question: there is a unique fundamental proposition, Pure is interpreted by the complete stock of denotations of closed terms containing only logical vocabulary, and both zeroary and unary QLN hold globally. The proof separates the two directions. Recombination is supplied by Bacon's generic-proposition construction. Exhaustion follows from cone invariance: if a closed logical unary operator is true of every proposition at a world, then its application to the fundamental proposition is true at every accessible world. At zeroary level, every closed logical proposition is necessarily equivalent to truth or falsity. This settles every nonvacuous QLN instance in the unique-fundamental-proposition case. It does not prove the broader extension Bacon mentions for arbitrary individual types or for several pairwise-distinct fundamental entities.

6.6 The countability error in M5

The notes propose choosing a pair $\{s, s'\}$ outside the orbit of r because the orbit is countable and, it is said, there are uncountably many world-indexed pairs. But Bacon's worlds are finite sequences of natural numbers, hence countable. The displayed family of pairs is therefore countable as well. Countability alone proves nothing.

The replacement is direct diagonalization. Enumerate the views of R by the worlds. Define a proposition s_R whose truth at the singleton $[n+2]$ disagrees with the corresponding singleton's membership in the n-th view of R, and let

$$s'_R = s_R \cup \{\langle\rangle\}.$$

Membership above a nonactual world depends only on the final symbol of its sequence. Hence every nonactual view of s_R and s'_R is empty or universal, while both members differ from every view of R. Exchanging their membership in the set T_0 of true propositions yields an invariant nonidentity operator $f_R = G_T$ with

$$f_R(R) = R.$$

If f_R and identity are both included among the pure operators, evaluation at R is not injective. This is exactly the pre-rebuild QSS obstruction the M5 strategy needs.

The separate rebuilding step is also now verified. Instead of retaining this old R, the rebuilt model chooses a new fundamental proposition generic for the enlarged application-closed stock containing Goodman's displayed f. That proposition again satisfies Recombination and fun'-separation.

6.7 The scope of several semantic claims

The M4 and M6 theorems are genuine mathematical results, but at present they are conditional. They assume that every operator counted as pure belongs to Bacon's function domain and is invariant, together with necessitated QSS and, for some results, the presence of identity and the two constant operators. The proofs do not yet instantiate these assumptions for Bacon's complete stock of operators denoted by closed terms containing only logical vocabulary. We therefore do not count them as unconditional verifications of the corresponding claims about Bacon's particular glued model.

7 New theorems advancing the consistency question

The audit was not merely defensive. The formalization has produced several theorems not stated in the original notes. These results materially clarify the logical relations among Goodman's central claims.

7.1 The exact footnote-59 obstruction

Theorem 1 (M1 for CEV+ Henkin models)**.** *No compositional CEV+ Henkin model validates full QLN,* PP*, and Purity of Fun. More generally, any such model that validates the PP-free M1 background,* QSS*, and unique proposition-level fundamentality must falsify* PP*.*

The proof has three machine-checked parts. First, Bacon's closed footnote-59 term is evaluated and shown to denote the operator D such that

$$Dp \quad \leftrightarrow \quad \forall X\, q\, (\mathsf{Pure}(X) \,\&\, \mathsf{Fun}(q) \,\&\, p = Xq \;\rightarrow\; {\sim}Xp).$$

Second, object-language QSS is evaluated using the same type-correct identity relation as the rest of the semantics. Third, QSS identifies each relevant pure operator X with D, yielding the diagonal contradiction. Thus the theorem does not assume a separate semantic diagonal premise.

The abstraction uses no feature of Bacon's tree representation. Its semantic assumptions are the ordinary compositional clauses for variables, constants, application, abstraction, conjunction, identity, Pure, and Fun, together with type closure, congruence of identity, global soundness of the CEV proof theory, and soundness of vector Equivalence. The exact tree theorem is an instance.

This is a model-independent inconsistency theorem for the stronger package, but it does not answer Goodman's stated question. Purity of Fun is deliberately absent from both of Goodman's central axiom sets, and the proof that D is pure uses it. Full QLN supplies QSS, so for the full-QLN variant Purity of Fun is the sole extra M1 premise. For the Recombination-only question, both QSS and Purity of Fun remain unavailable.

7.2 Recombination, Exhaustion, and the existence of a fun' proposition

Theorem 2. *Unary Recombination yields the pointwise* QSS *core. Adding zeroary Exhaustion yields the necessitated universal form needed for* QSS*. Together with unique proposition-level fundamentality, the resulting stock proves that a* fun' *proposition exists.*

This theorem removes the previously assumed existential fun' premise when T6 is applied to the repaired stock derived from a fundamental proposition. It does not alter Goodman's original T6 formulations, which take $\exists\mathsf{fun}'$ as a premise. It also makes explicit exactly where Exhaustion, rather than Recombination, enters the object-language argument.

7.3 An exact semantic consequence of T6

Theorem 3. *Any model of the repaired central stock must falsify global* $\mathsf{L2}$ *or global* TU*:*

$$\sim\mathsf{GValid}(\mathsf{L2}) \quad \vee \quad \sim\mathsf{GValid}(\mathsf{TU}).$$

In every such model, one of the two closed formulas is false at some world under some well-typed environment.

Here $\mathsf{GValid}(A)$ means that the closed formula A is true at every world of the model. The repaired central stock consists of the logical-purity schema, application closure, unique proposition-level fundamentality, the denial of fundamentals at every other type, zeroary and unary Recombination, PP, and zeroary Exhaustion. This formalizes the central dilemma in Goodman's notes. It does not select a disjunct. A positive model may exploit either cross-kind collisions on fun' inputs or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping.

7.4 The rebuilt M5 model

Theorem 4 (Completion of the rebuilding step)**.** *There is a typed interpretation of Bacon's Recombination background whose pure denotations form the least application-closed stock containing all closed logical denotations and Goodman's displayed exotic operator f. That stock is countable. The operator f is in the modalized unary function domain, commutes with taking views, is self-inverse, is not truth-uniform, and is not a biconditional operator. A rebuilt fundamental proposition supplies Recombination and* fun' *-separation at every world.*

This discharges the rebuilding step that Goodman's notes left "modulo Theorem 10.1." It verifies the intended M5 independence result for Bacon's Recombination background without relying on that unproved source theorem. It does not provide a model of PP: PP would add the further requirement that the classifier of this enlarged pure stock itself occur at the next type.

7.5 A uniform pre-rebuild M5 obstruction

Theorem 5 (Uniform version of the M5 construction)**.** *For every candidate proposition R, there is a pair $\{s_R, s'_R\}$ disjoint from the set of all views of R. Every nonactual view of either member is empty or universal. The associated invariant transposition fixes R and is not identity.*

This is stronger and cleaner than the original M5 selection argument. It shows that an exotic invariant operator fixing any preassigned R can be constructed without cardinal assumptions. It shows that the pre-rebuild failure is caused by keeping the old fundamental proposition fixed while enlarging the pure operators. The preceding theorem shows how rebuilding the fundamental proposition avoids that failure.

7.6 The T9 and M7 reductions

Theorem 6 (T9 dichotomy)**.** *If* PC *supplies an injection $\mathcal{P}(K) \to P$ into the pure unary operators, and every $\approx$-class injects into G, then*

$$K \textit{ is finite} \quad \vee \quad |G| \geq 2^{|K|}.$$

Theorem 7 (Pure Completeness, L2, and infinitely many kinds)**.** *Suppose that the type-correct Pure Completeness selector is realized by pure unary classifiers, the required instance of* L2 *identifies the represented kind at a* fun′ *proposition, and pure reversible operators act on kinds by the composition laws used in T9. Then K is infinite. Consequently the T9 counting inequality implies*

$$2^{|K|} \leq |G|.$$

The proof supplies the missing separation argument in the conditional infinitude route. If K had n members, classifiers of subsets of cardinalities $0, \ldots, n$ would have to occupy $n + 1$ different kinds. Pure reversible composition can identify two such classifiers only when the selected subsets have the same cardinality. This is impossible for a set of n kinds. The result is semantic and conditional: it does not derive L2 or the required classifier realization from the central stock.

Theorem 8 (Invariant reachability)**.** *Every proposition is reachable from r by an invariant operator if and only if*

$$i \longmapsto i \cdot r$$

is injective.

The second theorem turns the open invariant form of Fundamental Completeness into a precise property of the chosen fundamental proposition. It also confirms that the issue depends on Bacon's particular construction rather than following from general facts about substitutions.

8 Where the open question now stands

Goodman's question is whether the CEV+ axiom extension by the Recombination-only stock together with PP is consistent. A model of that stock would be a sufficient consistency certificate, but model existence is not the definition of the proof-theoretic question. Isabelle proves the exact finitary settlement standard:

$$\begin{aligned}&\text{the full axiom extension is consistent}\\ &\quad\Longleftrightarrow \text{every finite subset of its added axioms is consistent.}\end{aligned}$$

Equivalently, a negative answer requires and is guaranteed by a finite subset from which CEV+ derives falsity. No theorem presently establishes consistency for every finite subset or exhibits such a finite inconsistent core. The verified results clarify both possible answers without deciding between them.

Affirmative route. On the model-theoretic route, a positive answer would be established by a model of the full theory plus PP, with purity interpreted at every relevant type. T6 shows that such a model must falsify L2 or TU. Thus it must contain either pure operators from distinct $\approx$-classes that agree on suitable fun' propositions, or a pure reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping. Bacon's substitution-structure models do not already supply the desired example. The exact footnote-59 theorem now rules out every compositional CEV+ Henkin model that validates the PP-free M1 background, including Purity of Fun, together with QSS and unique fundamentality. This stronger package includes Purity of Fun, however, whereas Goodman's central question does not. Thus a positive model may evade the M1 argument by falsifying Purity of Fun; in the Recombination-only variant it may also evade QSS. M2 independently shows that identifying purity with invariance is incompatible with QSS. The operator Z shows that the operators denoted by Bacon's closed logical terms already take the $\neg$L2 branch. The missing step is construction of a Henkin interpretation that validates PP and the remaining background theory without importing Purity of Fun. Any enlarged pure stock must be checked afresh, since both fun' and right-cancellativity are relative to that stock.

Negative route. The most direct negative route is the one already isolated in Goodman's notes: derive enough of L2 and one of Inv, WI, TU, or RS from the remaining principles, and apply T6. The formalization verifies the relevant T6 implications. In the repaired stock it derives a fun' witness from Recombination, zeroary Exhaustion, and unique fundamentality; the original T6 formulations simply assume that witness. The exact Z counterexample shows that neither closed-logical purity nor merely adding further pure operators on Bacon's tree frame secures L2 under the stated closure and fun' conditions. Any negative proof deriving L2 from PP must therefore use features not present in these tree interpretations; alternatively, it would show that no such PP enlargement exists.

There is now a second, more direct negative route. The exact footnote-59 argument has been lifted from Bacon's tree construction to arbitrary compositional CEV+ Henkin models. The remaining issue on this route is no longer semantic generality. It is the use of Purity of Fun in certifying the diagonal operator as pure. Deriving Purity of Fun from Goodman's central stock, or replacing that step with a construction using only the stated principles, would yield a model-independent negative answer without first deriving L2.

Relative plausibility. The formal work does not establish that either answer is more likely. It shows exactly what each answer must overcome. A positive model must depart from the stronger M1 package by falsifying Purity of Fun, and in the relevant full-QLN setting must take one of the escape routes forced by the T6 disjunction. A negative proof may either eliminate the Purity-of-Fun premise from the exact M1 obstruction or promote one of Goodman's model-sensitive principles, especially L2 or TU, to a consequence of the background theory itself.

9 Goodman's open problems and suggested attacks

Goodman closes his notes with four proposed attacks, ordered by expected value. The formal work has not treated them evenly. The first has received a determinate answer in Bacon's model. The second now has exact sufficient granularity conditions and a sharp QLN reduction, but no derivation of those conditions from the central stock. The third now includes a conditional proof of infinitely many kinds and the resulting exponential lower bound on G. The fourth has been pursued through several model constructions and finite-fragment reductions, but no model of the complete theory with PP has been obtained.

9.1 Calibrate L2 in Bacon's model

This problem is settled negatively for Bacon's stock of operators denoted by closed terms containing only logical vocabulary. The immediate-successor variation operator

$$\widehat{Z}(P) = \{w : (\exists n\, P(w^\frown\langle n\rangle)) \,\&\, (\exists n\, \neg P(w^\frown\langle n\rangle))\}$$

is denoted by such a term. It is surjective and noninjective, is not reversible, and carries every fun' proposition to another fun' proposition. Identity and Z therefore give an explicit counterexample to L2. Thus both L2 and strong L2 fail in Bacon's exact appendix model. No corresponding theorem for arbitrary enlarged stocks is claimed.

This answers the model-theoretic calibration question. It does not decide whether L2 follows from PP in the full theory. We also made bounded automated searches for a derivation of L2 from the full stock and for several sufficient intermediate principles. No proof was found. Those failures provide no proof of underivability, so the object-language question remains open.

9.2 Cap or characterize G

The formalization verifies the four T6 routes through Inv, WI, TU, and RS, but it does not derive any of those principles from the central stock. The proposed extension of the M5 collision calculation to an existentially given reversible operator has been settled negatively. Every operator with a two-sided inverse is injective, while Goodman's rebuilt exotic operator is a reversible operator that is neither uniformly truth-preserving nor uniformly truth-flipping and nevertheless has no collision. Thus reversibility together with failure of truth-uniformity cannot by itself support a general collision argument.

The rebuilt M5 interpretation also shows that an enlarged pure stock can contain such an exotic reversible operator while validating Bacon's Recombination background. It thereby supplies the kind of failure of TU that T6 says a positive model may exploit. It does not validate PP, however: the classifier of the enlarged stock has not been shown to belong to the pure stock at the next type.

The granularity proposal now has an exact sufficient condition. If each pure reversible operator preserves sameness of truth value, in the sense that propositions with the same truth value are sent to propositions with the same truth value, then surjectivity forces that operator either to preserve truth everywhere or to reverse truth everywhere. Thus this same-truth-value preservation implies TU.

The condition is substantive. The Fregean principle that propositions with the same truth value are identical implies it, but that principle leaves only truth and falsity and is incompatible with the required fundamental proposition distinct from both. At the other extreme, Isabelle verifies a four-proposition bijection that sends two true propositions to propositions with different truth values and is neither uniformly truth-preserving nor uniformly truth-flipping. Thus invertibility and ordinary functionality alone do not supply the needed granularity.

Full unary QLN reduces the remaining issue to one displayed condition. We have now encoded this reduction in the complete object language. For a pure reversible Z, let $Zr \leftrightarrow r$ say that Z agrees with the identity operator at the fundamental proposition. The two directions of QLN for this agreement operator and its negation yield

$$\begin{aligned}&\Box(Zr \leftrightarrow r) \vee \Box{\sim}(Zr \leftrightarrow r)\\ &\quad \iff (Z \text{ preserves truth everywhere})\\ &\qquad\qquad \vee (Z \text{ reverses truth everywhere}).\end{aligned}$$

The two operator-builders are closed terms containing only logical vocabulary. Isabelle verifies from the logical-purity schema and application closure that their values at Z are pure whenever Z is pure. It also verifies the required unary Exhaustion instance and proves in CEV+ that the displayed modal disjunction is equivalent to

$$\forall p\,(Zp \leftrightarrow p) \vee \forall p\,{\sim}(Zp \leftrightarrow p).$$

The verified truth-condition calculation identifies these two disjuncts with uniform truth-preservation and uniform truth-reversal, respectively. Consequently Attack 2 is reduced to proving, for each $Z \in G$, that the agreement proposition is either necessary or necessarily false. More importantly, the test shows that this claim is not a weaker intermediate granularity principle: in the full QLN setting it is exactly the corresponding instance of TU. The purity proof for the two constructed operators uses the logical-purity schema, application closure, and the assumed purity of Z, not PP. The latter is present in the full-QLN-plus-PP stock used for the corollary, but the reduction gives no proof of either disjunct from it; hence we have not derived TU from that stock. Because the equivalence uses unary Exhaustion, it does not by itself reduce TU in Goodman's Recombination-only theory.

This conclusion is proof-theoretic. It identifies what follows in CEV+ from full QLN, the logical-purity schema, application closure, and the displayed local assumptions. It does not prove that TU is underivable from PP. Such a negative proof-theoretic claim would require a completeness argument or an appropriate countermodel. The rebuilt M5 interpretation is not that countermodel, since it does not validate PP. Its role is only model-theoretic calibration: reversibility and the PP-free Recombination background permit a failure of truth uniformity.

There is also a cardinal form of the granularity route. Once Attack 3 gives $2^{|K|} \leq |G|$, any injection of G into a stock of descriptions of cardinality at most $|K|$ contradicts Cantor's theorem. In particular, if K is infinite, descriptions by finite strings over K provide such a ceiling. Bacon's stock of operators denoted by closed logical terms is countable, so it cannot realize the complete Pure Completeness–L2 package that yields the exponential lower bound. This does not refute PP: the Purity schema says that closed logical terms denote pure entities, not that every pure entity is denoted by a closed logical term. A model of PP may therefore require a larger, even uncountable, pure stock.

9.3 Attack 3's infinitude route

The formalization verifies the starting point, the missing conditional infinitude step, and the counting conclusion of this proposal. The five kinds represented by identity, necessity, possibility, constant truth, and constant falsity are pairwise distinct. Under L2, the T8 construction gives 31 pairwise-distinct propositions and 31 pairwise-distinct pure unary operators. More importantly, the type-correct Pure Completeness construction supplies a classifier for each set of kinds. If two classifiers belong to the same kind, composition by a pure reversible operator carries one selected set to the other and hence preserves its cardinality. If K had n members, the classifiers of subsets having cardinalities $0, \dots, n$ would therefore occupy $n + 1$ different kinds. Hence K is infinite.

The T9 theorem then proves, under its stated representation hypotheses, that

$$|G| \geq 2^{|K|}.$$

This proof does not obtain infinitely many kinds by iterating T8, and it does not use Goodman's earlier premise refuted in T5. It instead uses the entire Pure Completeness selector and L2 to classify represented inputs, followed by the action of pure reversible composition on kinds. These are additional semantic premises. We have not derived the required global L2 or the classifier realization from the Recombination-only CEV+ stock with PP. Thus the conditional infinitude route is complete, while its application to Goodman's central theory remains open.

The combination with Attack 2 is now exact. If G has cardinality at most $|K|$, then $2^{|K|} \leq |G| \leq |K|$, contrary to Cantor's theorem. The same contradiction follows if each member of G has a unique description in a stock no larger than K, or if G is countable. The open task is therefore no longer to find the cardinal argument. It is to justify such a ceiling for the complete pure reversible stock, or to derive the agreement noncontingency condition above and obtain TU directly.

9.4 Prove consistency

This proposal has received the broadest constructive investigation. We have verified models for successively richer explicitly generated stocks of pure operators, including the Boolean operators, necessity and possibility, and the six displayed higher-order quantified operators. We have also rebuilt Goodman's M5 interpretation with its exotic reversible operator and a new fundamental proposition satisfying Recombination. These constructions show separately that the failures of L2 and TU required by T6 are genuine semantic possibilities. Neither construction supplies the missing self-classification required by PP.

At the proof-theoretic level, Isabelle proves that the complete Recombination-only extension with PP is consistent if and only if every finite subset of its added axioms is consistent. We have constructed tailored models for the first two classifier-bearing cyclic packages. For acyclic finite application dependencies, Isabelle proves the more limited stabilization result that the generated unary values are independent of the proposed classifier; this is not itself a general model theorem. The two cyclic models show that classifier-bearing cyclicity and its first nontrivial enlargement do not produce an inconsistent core. These results do not cover every finite package, since later applications may enlarge the very stock whose classifier must itself be pure.

Accordingly, the positive route remains open at a precise point. It requires a Henkin interpretation of the complete stock in which the interpretation of Pure has its own denotation in the appropriate higher-type pure stock, while the remaining background principles continue to hold. Equivalently, the finite-fragment route requires a tailored model for every remaining classifier-bearing finite package. No such general construction, and no finite inconsistent fragment, is presently known.

Assessment. The first suggested attack is complete in its original semantic form. The second now has exact sufficient conditions, a QLN equivalence, and a cardinal ceiling theorem, but none of the decisive premises has been derived from the central stock. The third is complete as a conditional semantic theorem: Pure Completeness plus L2 and the ordinary composition assumptions on pure reversible operators and kinds imply infinitely many kinds and hence the T9 exponential lower bound. The fourth has produced the strongest reductions and partial models, but the self-classification required by PP remains the central obstruction.

10 Audit of the error-prone zones identified in the notes

Goodman's final section lists five recurrent sources of error. The present formalization does not assume them away; it addresses each one explicitly.

1. **Inner and outer negation.** The T6 encodings distinguish composition with negation on the input from negation of the output, and the four contradiction routes were audited clause by clause. In particular, the truth-flipping route uses the correctly oriented conjugate operator. No result in this report relies on interchanging these two operations.

2. **Truth at a world and identity of propositions.** The semantic evaluator keeps these notions separate. Identity is interpreted at each type by its proper local equivalence relation, while truth is evaluation at a world. The exact-carrier M1 calculation, the secondary rebuilt M5 model, and the exact-carrier L2 counterexample all use this distinction. Higher-type identity is interpreted pointwise rather than by treating operators themselves as propositions.

3. **Validity in Bacon's model and derivability in CEV+.** Object-language derivations and semantic theorems occur in separate Isabelle theories and are reported separately here. The conditional M4 and M6 stock theorems are not promoted to claims about Bacon's appendix model, and the rebuilt M5 interpretation is not promoted to a model of PP.

4. **Group and monoid reasoning.** Bacon's substitutions are treated as a monoid; no cancellation or group inverse is assumed. The M7 analysis proves the exact condition for invariant reachability—injectivity of $i \mapsto i \cdot r$— and leaves its instance open. It does not infer injectivity from a stabilizer claim.

5. **Cardinality and reachability.** The false countability selection in M5 is explicitly refuted and replaced by a uniform diagonal construction. The M6 and M7 results keep the countability of substitution orbits separate from the injectivity property expressed by fun'; nowhere is fun' treated as a surjectivity or reachability principle.

Thus the development encountered two of the dangers Goodman flags—the M5 countability argument and the possible conflation of model validity with derivability—but the statements retained in this report incorporate the corrections. The remaining three are enforced by the typed encodings and the explicit hypotheses of the checked theorems.

11 Remaining verification work on the notes

The following items should not be reported as verified:

1. T7b, until a precise kind-level diagonal formula is supplied.
2. The multiple-fundamental "wide Fun" discussion in M4, which does not state a precise selection condition on Fun or a precise Separated Structure formula.
3. Bacon's suggested extension of the appendix construction to arbitrary individual types or to several pairwise-distinct fundamental entities. The QLN verification is complete for the unique-fundamental-proposition specialization used in Goodman's question, but not for that broader future-work claim.

12 Recommended next steps

The following tasks arise directly from the unresolved or qualified claims in Goodman's notes.

1. **Derive or refute TU in the full-QLN-plus-PP theory.** The QLN granularity test is complete. The agreement and disagreement operators have been formalized in the complete object language, their purity follows from the logical-purity schema and application closure, and full QLN proves the equivalence between

$$\Box(Zr \leftrightarrow r) \vee \Box{\sim}(Zr \leftrightarrow r)$$

and a pointwise disjunction that the verified truth-condition lemma identifies with truth-uniformity. Thus the displayed claim is exactly the remaining TU premise, not a weaker lemma. A useful next step must exploit a genuinely PP-specific constraint on pure reversible operators, or construct a model of the full-QLN-plus-PP stock in which one such operator is not truth-uniform. For Goodman's Recombination-only question, the parallel task is either to obtain the required noncontingency without unary Exhaustion or to use a different route to TU.

2. **Discharge or refute the conditional infinitude package.** The cardinal reasoning no longer contains a gap. What remains is to derive from the object theory the type-correct classifier realization and the required instance of L2 used in the infinitude theorem, or to construct a model showing that one of them can fail with PP. If they hold, determine whether the complete pure reversible stock has descriptions of cardinality at most $|K|$. Such a ceiling immediately contradicts the verified exponential lower bound.

3. **Settle the finite-fragment alternatives.** An affirmative answer requires a consistency proof for every finite subset of the Recombination-only PP stock. A negative answer requires one finite subset from which CEV+ derives falsity. The compactness theorem makes these exhaustive proof-theoretic targets; arbitrary finite packages remain untreated.

4. **Eliminate the Purity-of-Fun premise from the M1 route.** The type-correct footnote-59 proof is now established for arbitrary compositional CEV+ Henkin models. Its remaining extra premise is not semantic: the object-language proof that the diagonal is pure uses Purity of Fun, which Goodman's central axiom sets omit. Determine whether that principle follows from the stated stock or whether the construction can be replaced by one requiring only the stated principles.

5. **Obtain determinate formulations.** Obtain an exact T7b statement, an exact formulation of the wide-Fun discussion, and then encode them.

6. **Isolate any genuinely PP-specific M5 premise.** The unrestricted existential generalization of the M5 collision method is false: reversible operators are injective, and the rebuilt exotic operator is a non-truth-uniform bijection. Further work should pursue this route only if PP supplies an additional constraint on the inverse witness that is strong enough to force the displayed kind of collision.

7. **Test Goodman's negative route.** Determine whether L2 follows from PP in the full theory even though it fails when the pure unary operators are precisely Bacon's closed

logical denotations. A positive derivation, together with TU, WI, Inv, or the strong-L2+RS route, would yield the corresponding T6 contradiction and would also explain why Bacon's exact interpretation cannot validate PP. The bounded Vampire experiments show that further work should concentrate on the agreement noncontingency condition and the hypotheses of the conditional infinitude theorem, rather than repeating the monolithic refutation search.

13 Reproducibility

The repository's background-theory organization, Goodman project index, and current build commands are documented in `README.md` and `docs/REPOSITORY_STRUCTURE.md`. We do not reproduce that implementation documentation here.

The controlling verification ledger is

`reports/GOODMAN_COMPLETE_VERIFICATION_MATRIX_2026-07-27.md`.

The dedicated audit theory is

`reports/audit_goodman_complete/Audit_Goodman_Complete.thy`.

The verification matrix gives, for each claim, the exact Isabelle theorem and the qualification under which it has been proved, and identifies the source theory corresponding to each result. `STATUS.md` gives the concise current project status. The audit theory is a kernel-cleanliness gate. It checks that its 167 listed theorems have no oracles, dangling hypotheses, or unsolved unification pairs; the source-fidelity and hypothesis-strength assessments reported here come from the separate statement-by-statement audit. The repository's serial check also builds the maintained auxiliary model and finite-fragment sessions used in the progress sections of this report.

The appropriate overall verdict is therefore: *the formalization verifies, corrects, or precisely qualifies every determinate claim in Goodman's notes. It completes the rebuilding step in M5, proves that the unrestricted existential generalization of the M5 collision method is false, refutes L2 when the pure unary operators are precisely the closed logical denotations in Bacon's appendix model, proves the exact T6 semantic consequence, and completes Bacon's footnote-59 denotation argument in the exact recursively restricted semantics and in arbitrary compositional CEV+ Henkin models. The last result proves that full QLN, PP, and Purity of Fun have no such model, but it does not settle Goodman's question because the central axiom sets* *omit Purity of Fun. The report also corrects the use of* PC *as a surrogate for Bacon's stronger Melianism premise, the attribution of the infinitary argument to note 60, and the omission in Bacon's text of the QLN-validation proof, which is now supplied for the specialization relevant here. The scopes of QLN and Modalized Functionality are now stated exactly, and the remaining underspecified and model-dependent claims are explicitly identified. The formalization also proves the conditional core of Goodman's third suggested attack: a type-correct Pure Completeness realization, the required instance of L2, and the standard action of pure reversible operators imply infinitely many kinds, after which T9 forces* $2^{|K|} \leq |G|$*. The second suggested attack is reduced both to the noncontingency of* $Zr \leftrightarrow r$ *under QLN and to a cardinal ceiling on the descriptions of* G*. These results rule out the full Pure-Completeness–L2 package in Bacon's countable closed-logical stock, but they do not rule out a larger interpretation of Pure and therefore do not settle PP. The exact compactness theorem reduces the open question to consistency of every finite added-axiom fragment, or equivalently to the absence of a finite inconsistent core; neither side has been established. Goodman's consistency question itself remains open.*